FUN-3.F.1 Differentiating f' produces the second derivative f'', provided the derivative of f' exists; repeating this process produces higher order derivatives of f.
FUN-3.F.2 Higher-order derivatives are represented with a variety of notations. For $y = f(x)$ notations for the second derivative include $\frac{d^2y}{dx^2}$, $f''(x)$, and y''. Higher-order derivatives can be denoted $\frac{d^ny}{dx^n}$ or $f^{(n)}(x)$.

3-5 & 3-6 Derivative Procedures and Higher Order Derivatives

During the course of the year, you will need to use the properties of logarithms you learned in Advanced Algebra:

Logarithmic Properties	➢ **Power Property:** $\log_b(x^k) = k \cdot \log_b x$ ➢ **Product Property:** $\log_b(xy) = \log_b x + \log_b y$ ➢ **Quotient Property:** $\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$ ➢ $\text{Log}_b(b^x) = x$ ➢ $b^{\log_b x} = x$

We have two more derivative formulas:

Derivative of Logarithmic functions	WITHOUT THE CHAIN RULE:	$\frac{d}{dx}(\log_a x) = \frac{1}{x(\ln a)} = \frac{1}{x} \cdot \frac{1}{\ln a}$
	WITH THE CHAIN RULE:	$\frac{d}{dx}(\log_a u) = \frac{1}{u(\ln a)} \cdot \frac{du}{dx} = \frac{1}{u} \cdot \frac{1}{\ln a} \frac{du}{dx}$

Derivative of Exponential functions	WITHOUT THE CHAIN RULE:	$\frac{d}{dx}(b^x) = (\ln b)b^x$
	WITH THE CHAIN RULE:	$\frac{d}{dx}(b^u) = (\ln b)b^u \cdot \frac{du}{dx}$

What derivative formulas do these formulas remind you of?

Example 1: Differentiate $y = \log_4(x^3 + 2x)$

$$y' = \frac{1}{x^3+2x} \cdot \frac{1}{\ln 4} \cdot (3x^2+2)$$

$$= \frac{3x^2+2}{(x^3+2x)\ln 4}$$

Example 2: Differentiate: $y = 8^{9x^2}$

$$y' = 8^{9x^2} \cdot (\ln 8) \cdot 18x$$

Higher order derivatives use similar notation to the first derivative:

Function	y	$f(x)$
1st derivative	y' or $\frac{dy}{dx}$	$f'(x)$
2nd derivative	y'' or $\frac{d^2y}{dx^2}$	$f''(x)$
3rd derivative	y''' or $\frac{d^3y}{dx^3}$	$f'''(x)$
*n*th derivative	$y^{(n)}$ or $\frac{d^{(n)}y}{dx^{(n)}}$	$f^{(n)}(x)$

Example 3: Let $y = e^{4x^2}$. Find $\frac{d^2y}{dx^2}$

$$\frac{dy}{dx} = e^{4x^2} \cdot 8x$$

$$\frac{dy}{dx} = \underline{8x e^{4x^2}}$$

$$\frac{d^2y}{dx^2} = 8 \cdot e^{4x^2} + 8x \cdot e^{4x^2} \cdot 8x$$

$$= 8e^{4x^2} + 64x^2 e^{4x^2}$$

$$= 8e^{4x^2}(1 + 8x^2)$$

Example 4: Find $\frac{d^2y}{dx^2}$ for $\boxed{\sin y = x + y}$

$$\cos y \cdot \frac{dy}{dx} = 1 + \frac{dy}{dx}$$

$$\cos y \cdot \frac{dy}{dx} - \frac{dy}{dx} = 1$$

$$\frac{dy}{dx}(\cos y - 1) = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y - 1} \quad \text{or} \quad (\cos y - 1)^{-1}$$

$$\frac{d^2y}{dx^2} = -1(\cos y - 1)^{-2} \cdot \left(-\sin y \cdot \frac{dy}{dx}\right)$$

$$= \frac{-1}{(\cos y - 1)^2} \cdot \left(-\sin y \cdot \frac{1}{(\cos y - 1)}\right)$$

$$= \frac{\sin y}{(\cos y - 1)^3}$$